

William F. CHAPARRO-PICO

Microbiologist (BSc, MSc in Microbiology)

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EDUCATION

Master's degree in Microbiology

2023 – 2025

Industrial University of Santander (Bucaramanga, Colombia)

School of Microbiology

Molecular epidemiology of Respiratory Syncytial Virus and Rhinovirus in population of Santander during the years 2020 – 2024.

Bachelor's degree in Microbiology and Bioanalysis

2017 – 2022

Industrial University of Santander (Bucaramanga, Colombia)

School of Microbiology

Evaluation of the efficacy of physical, chemical and physicochemical treatments to saliva samples for the diagnosis of SARS-CoV-2 by direct RT-qPCR.

COMPLEMENTARY ACADEMIC TRAINING

COURSES

Portable sequencing techniques (MinION) applied to environmental monitoring

Alexander Von Humboldt Institute and Ecopetrol, Colombia

2025

Theoretical and Practical Introductory Course on Flow Cytometry and Operation of the Attune NxT Instrument

ThermoFisher, Colombia

2023

Pathobiology of Intracellular Infections

Yale University, Fogarty International Center, International Center for Medical Training and Research (CIDEIM, Colombia)

2023

IX International Course on Trypanosomatids

Universidad de Antioquia, Colombia

2022

High content and high predictive cellular models for host-pathogen interactions studies, disease modeling and drug discovery

Institut Pasteur, Montevideo, Uruguay

2022

PROFESSIONAL SKILLS

Laboratory skills

BSL-1 and BSL-2 laboratory operations; bacteria isolation; Sanger and Nanopore sequencing; DNA/RNA extraction and purification; PCR, RT-PCR, RT-qPCR; gel electrophoresis for nucleic acids and proteins; restriction enzyme digestion; Western blotting; ELISA; flow cytometry; cell culture; primer design; multiple sequence alignment; phylogenetic analysis.

Software

Metagenomic analysis tools (Kraken2, Bracken, NanoPlot, NanoFilt); Python; R; IQ-TREE; BEAST 2; Tracer; FigTree; Office; L^AT_EX.

Languages

English (B2), Spanish (Native).

EMPLOYMENT

Microbiologist — Molecular biology analyst — DNA sequencing by NGS — PSL Proanalysis S.A.S.

July 2025 – Present

PSL Proanalysis – Ecopetrol

Piedecuesta, Colombia

Microbiologist in charge of molecular and bioinformatic analyses of oil-contaminated soil samples using metagenomic approaches. Responsibilities include DNA extraction, 16S rRNA gene amplification by PCR, Oxford Nanopore sequencing, and taxonomic profiling through bioinformatic pipelines.

Adjunct Professor (non-tenured) – General Biology

February 2025 – June 2025

Universidad Industrial de Santander, School of Microbiology

Bucaramanga, Colombia

Adjunct professor for first-semester undergraduate students in Microbiology and Bioanalysis. Duties included lecturing on common laboratory practices in biology.

Adjunct Professor (non-tenured) – Immunology

February 2025 – June 2025; February 2024 – June 2024

Universidad Industrial de Santander, School of Microbiology

Bucaramanga, Colombia

Adjunct professor for third-semester undergraduate students in Microbiology and Bioanalysis. Duties included lecturing on common laboratory practices in immunology (clinical diagnostic and research) and advising research seminars on immune responses against bacterial, viral and fungal pathogens.

Adjunct Professor (non-tenured) – Molecular Biology

August 2024 – December 2024; January 2023 – February 2023

Universidad Industrial de Santander, School of Microbiology

Bucaramanga, Colombia

Adjunct professor for second-semester undergraduate students in Microbiology and Bioanalysis. Duties included lecturing on common laboratory practices in molecular biology.

Research assistant

March 2023 – May 2023

Universidad Industrial de Santander, School of Microbiology

Viral DNA and RNA extraction, standardization and RT-qPCR performance for the detection of Respiratory Syncytial Virus and Rhinovirus in respiratory tract samples from patients with acute respiratory infection.

Research assistant

August 2022 – December 2022

Universidad Industrial de Santander, School of Microbiology

Performance of molecular biology and immunology tests for the diagnosis of infectious diseases (*Trypanosoma cruzi*). Report writing, administrative management and database updates for the Immunology and Molecular Epidemiology Research Group (GIEM-UIS).

PUBLICATIONS

Carbone-Schellman, J., Fontecilla-Escobar, J., Sales-Salinas, N., **Chaparro-Pico, W. F.**, Molina-Berríos, A., Ruete, M. C., González, P. A., Machuca, M. A., Opazo, M. C., Ezquer, M. E., and Duarte, L. F. (2025). **Therapeutic challenges in central nervous system viral infections: Advancing mesenchymal stem cell-based strategies for treating neuroinflammation and promoting tissue repair.** *Frontiers in Immunology*, 16. doi: 10.3389/fimmu.2025.1677433.

Sosa, L. M., Díaz, M. L., Lozano-Parra, A., Bueno, N., Niño, D., Herrera, V., Machuca, M. A., **Chaparro-Pico, W. F.**, & Niederbacher, J. (2025). **Virus respiratorios en pacientes con síntomas de SARS-CoV-2 durante el periodo pandémico en Santander, Colombia.** *Salud UIS*, 57, e25v57a19. doi: 10.18273/saluduis.57.e:25v57a19.

Chaparro-Pico, W. F., et al. (In preparation). **Impact of COVID-19 Non-Pharmaceutical Interventions on RSV and RV circulation and evolution in Colombia.**

SCIENTIFIC EVENTS

POSTERS

Molecular characterization of Respiratory Syncytial Virus and Rhinovirus identified from respiratory samples from the population of Santander during the period 2020–2024 (Spanish: Caracterización molecular de Virus Sincitial Respiratorio y Rinovirus identificados a partir de muestras respiratorias de población de Santander durante el periodo 2020–2024).

October 2024

XXVI Latin American Congress of Clinical Biochemistry and XXII International Congress of the National College of Bacteriology (Spanish: XXVI Congreso Latinoamericano de Bioquímica Clínica y XXII Congreso Internacional del Colegio Nacional de Bacteriología).

Differential circulation of Rhinovirus species and genotypes/clades during the COVID-19 pandemic and post-pandemic period (Spanish: Circulación diferencial de especies y genotipos/clados de Rinovirus durante la pandemia de COVID-19 y en el periodo pospandemia).

September 2024

XIV National Meeting on Research and Innovation in Infectious Diseases and IV Latin American Meeting on Research in Infectious Diseases (Spanish: XIV Encuentro nacional de investigación e innovación en enfermedades infecciosas y IV encuentro latinoamericano de investigación en enfermedades infecciosas).

Detection of respiratory viruses during the introduction and the first wave of SARS-CoV-2.

November 2023

XX Latin American Congress of Pediatric Infectious Diseases (Spanish: XX Congreso Latinoamericano de Infectología Pediátrica).

Detection of viruses causing acute respiratory infection during the SARS-CoV-2 pandemic period (2020–2021) (Spanish: Detección de virus causantes de infección respiratoria aguda durante el periodo pandémico (2020–2021) de SARS-CoV-2).

November 2023

U23 Fest International Congress: United for our well-being (Spanish: Congreso internacional U23 Fest: Unidos por nuestro bienestar).

Chagas disease in population assisted in an elderly care center in Bucaramanga (Spanish: Enfermedad de Chagas en población asistida en un centro de atención del adulto mayor en Bucaramanga).

November 2023

U23 Fest International Congress: United for our well-being (Spanish: Congreso internacional U23 Fest: Unidos por nuestro bienestar).

Replication study of genetic markers associated with Chagas disease in Latin American population (Spanish: Estudio de replicación de marcadores genéticos asociados a la Enfermedad de Chagas en población latinoamericana).

June 2022

IX Symposium on Molecular Biology of Chagas Disease (Spanish: IX Simposio de Biología Molecular de la Enfermedad de Chagas).

ORAL PRESENTATIONS

Molecular epidemiology of Rhinovirus infection: A66 genotype and fatal outcome in a case of acute respiratory infection (Spanish: Epidemiología Molecular de la infección por Rinovirus: Genotipo A66 y desenlace fatal en un caso de infección respiratoria aguda).

September 2024

4th International Health Science Meeting: Science Transforms (Spanish: IV encuentro internacional de ciencias de la salud: la ciencia transforma).

Validation of a rapid immunochromatographic assay for the diagnosis of Chagas disease in the population of Santander (Spanish: Validación de un ensayo inmunocromatográfico rápido para el diagnóstico de la enfermedad de Chagas en población santandereana).

November 2023

U23 Fest International Congress: United for our well-being (Spanish: Congreso internacional U23 Fest: Unidos por nuestro bienestar).

Association between genetic variants of the Masp2 gene and susceptibility to infection by *Trypanosoma cruzi* (Spanish: Asociación de variantes genéticas del gen Masp2 y susceptibilidad a la infección por *Trypanosoma cruzi*).

August 2023

VIII Colombian Meeting on Leishmaniasis and Chagas disease (Spanish: VIII Reunión Colombiana Leishmaniasis y enfermedad de Chagas).

Single nucleotide variant of the ITGAM gene is associated with the development of Chronic Chagasic Cardiomyopathy (Spanish: Variante de un solo nucleótido del gen ITGAM está asociada con el desarrollo de Cardiomiopatía Chagásica Crónica).

August 2023

VIII Colombian Meeting on Leishmaniasis and Chagas disease (Spanish: VIII Reunión Colombiana Leishmaniasis y enfermedad de Chagas).

Evaluation of the effectiveness of physical, chemical and physicochemical treatments for the diagnosis of SARS-CoV-2 by direct RT-qPCR (Spanish: Evaluación de la efectividad de tratamientos físico, químicos y fisicoquímicos para el diagnóstico de SARS-CoV-2 mediante RT-qPCR directa).

December 2022

XVIII Congress of the Colombian Association of Parasitology and Tropical Medicine (Spanish: XVIII Congreso de la Asociación Colombiana de Parasitología y Medicina Tropical).

Seroprevalence of *Trypanosoma cruzi* infection and characterization of patients with chronic Chagasic cardiomyopathy from five municipalities in Colombia (Spanish: Seroprevalencia de la infección por *Trypanosoma cruzi* y caracterización de pacientes con Cardiomiopatía Chagásica Crónica procedentes de cinco municipios de Colombia).

November 2022

III International Meeting of Health Sciences and I International Congress on Life, Health and Well-being (Spanish: III Encuentro Internacional de Ciencias de la Salud y I Congreso Internacional Vida, Salud y Bienestar).

Identification of genetic markers associated with Chagas disease in Latin American population (Spanish: Identificación de marcadores genéticos asociados a la Enfermedad de Chagas en población latinoamericana).

November 2021

U21 International Congress – Sustainable Development Goals (Spanish: Congreso Internacional U21 – Objetivos de desarrollo sostenible).

RESEARCH EXPERIENCE

Molecular characterization of Respiratory Syncytial Virus and Rhinovirus in population of Santander during the period 2020–2024

2023 – 2024

Universidad Industrial de Santander

Responsibilities: RNA extraction from clinical samples; RT-qPCR and RT-PCR amplification; Sanger sequencing; phylogenetic analysis; statistical analysis of epidemiological data.

Chagas disease cardiomyopathy: biomarkers of cardiac tissue

2024

Universidad Industrial de Santander – Johns Hopkins University – Fundación Cardiovascular de Colombia

Responsibilities: dissection of cardiac tissue from explants of heart transplant patients; isolation of RNA, proteins and inflammatory infiltrate cells.

Determination of the effectiveness of three different methods for pretreatment of human saliva samples used in the diagnosis of SARS-CoV-2 by direct RT-qPCR

2021 – 2022

Universidad Industrial de Santander

Responsibilities: collect saliva samples; apply physical, chemical and physicochemical pretreatments; RT-qPCR detection; comparison with gold standard; statistical analysis.

Molecular identification of *Trypanosoma cruzi* in peripheral blood samples from patients in the chronic phase of Chagas disease

2019 – 2020

Universidad Industrial de Santander

Responsibilities: DNA extraction; PCR; electrophoresis; statistical analysis and interpretation of results.

GRANTS AND FELLOWSHIPS

Master's Scholarship

High-level human capital development for the Department of Santander at the Universidad Industrial de Santander (BPIN 2020000100536).